Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
Faculty: Prof. Karthika V	**Slot:** L43 + L44
Regno: 23BCE1145	**Name:** S Vishwajith

EX4 – RSA

Code:

```
import random


def nBitRandom(n):
    return random.randrange(2**(n-1) + 1, 2**n - 1)


first_primes_list = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83,
89, 97]


def getLowLevelPrime(n):
    while True:
        pc = nBitRandom(n)
        for divisor in first_primes_list:
            if pc % divisor == 0 and divisor**2 <= pc:
                break
        else:
            return pc


def isMillerRabinPassed(mrc):
    maxDivisionsByTwo = 0
    ec = mrc - 1
    while ec % 2 == 0:
        ec >>= 1
        maxDivisionsByTwo += 1
    assert 2**maxDivisionsByTwo * ec == mrc - 1
```

```
    def trialComposite(round_tester):
        if pow(round_tester, ec, mrc) == 1:
            return False
        for i in range(maxDivisionsByTwo):
            if pow(round_tester, 2**i * ec, mrc) == mrc - 1:
                return False
        return True

    numberOfRabinTrials = 20
    for _ in range(numberOfRabinTrials):
        round_tester = random.randrange(2, mrc)
        if trialComposite(round_tester):
            return False
    return True

def generateLargePrime(n=1024):
    while True:
        prime_candidate = getLowLevelPrime(n)
        if isMillerRabinPassed(prime_candidate):
            return prime_candidate

P = generateLargePrime(1024)
Q = generateLargePrime(1024)
n = P * Q
phi_n = (P - 1) * (Q - 1)
e = 65537
d = pow(e, -1, phi_n)
M = int(input("Enter the message (a number): "))
C = pow(M, e, n)
D = pow(C, d, n)
```

```
print(f"\nValue of p is: {P}.")

print(f"Value of q is: {Q}.")

print(f"Value of n is: {n}.")

print(f"Value of phi(n) is: {phi_n}.")

print(f"Value of e is: {e}.")

print(f"Value of d is: {d}.")

print(f"Encrypted text is: {C}.")

print(f"Decrypted text is: {D}.")
```

Output:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-7> python .\RSA_Chipper.py
Enter the message (a number): 1907494793475927

Value of p is: 100131935517107580263233761267815647946558155749052343146815555125392377893290118289978735965296608360267866697692855099026605166698752183691
954517982452242564033414898844227407418045122308546919916369434790973967502346646818058798352240050072778446328090908017717370648755971138110003525551861999
498141826331.
Value of q is: 102832499015477493950502783925233354541328959799065314954111349613958084651330804228367085612826619750499364408846355699574821774346500820806
486574709408512888726764690240131155954232072407779410824274400241540071056532844567851087048358174581072377430145660568955813403703188443715948699701873577
592771437311.
Value of n is: 10296817160480821156529572914942735792844131447424874791147176428671299609815632612819058522488531237132790734354127156962695375611572676262 2
392539136530611276883398866521183884103287858026472681939710307684041906681821739216570548271435010210014495202353176873891066704999008913158532885134678508
257342121412810848995793521202054325060007853489308646297759640705765730941498492918262858409421584357153889077392573953530525542143078157856036980845202837
454912917374406887270927638358467743197841477049015461415907156888033108928140931821936568123518559048127039486610411328418049011445789509537840876470549534
15635941.
Value of phi(n) is: 10296817160480821156529572914942735792844131447424874791147176428671299609815632612819058522488531237132790734354127156962695375611572676 2
626223925391365306112768833988665211838841032878580264726819397103076840419066818217392165705482714350102100144952023531768738910667049990089131585328851346
785082573421214107812046504676704599169596080773634644297751422784641247564618941049882928166491842381261413726567962849016424651334341570638884455844508002
184265279943098198792854750367947728340204258943137857160549775568317476475201332270727230825621412720205398894571209245445965775244198499712500155883391147
7862502372300.
Value of e is: 65537.
Value of d is: 9061111117067456821938348248326555360253089839874992431070081923101930071521234784114328442085517700046116044855138645289722869558732004598 86
571328727587990533655702798420696242428676525947591820319357441560349854609460815127036126144040756951333745412532220985251933871050374908781132574192773384
228364866564318420831251276494854951327805316953872467767136668294696838481594043795614964333640723686656981972831049258319601967545995174395188435013839067
202112979833083319638726525190180146482472018043214721861796786632028896825224184519525996583828977306941845229101169716926162067471434751499002196229231252
9362273.
Encrypted text is: 433902901355408644361496003587444562179202062555353314179850849480778459306574497318018360627753839224786089878482044357922467512003194 70
784319733400075428322825535623705435094834956438057370143986932064410275508880311251479212295713832619639604905062628288112896800388773254149133256107104723
773604418069356987356941666524763050877094656459184705919839539753264467721648145042932903011953114250324966936124119053905438358883210742420475028704082724
119018696200144549882595308147787919854472016336039098200625163007109629524517329340528689062868626502125371444632192775833914235056672528207631664818655020 58
15873698437.
Decrypted text is: 1907494793475927.
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-7>
```